

Multiple Choice

1. **Answer:** A

Options: A) a neutron; B) a proton; C) an electron; D) an atom

Note: A neutron always has a mass of approximately one atomic mass unit (amu) and carries no electrical charge, distinguishing it from protons and electrons.

2. **Answer:** B

Options: A) The winner experienced less water resistance.; B) There was more gravitational pull on the winner.; C) Friction slowed down the second place swimmer.; D) There was more force in the winner's kicks and strokes.

Note: The statement about gravitational pull is incorrect because the differences in swimming speed are primarily influenced by factors such as technique, strength, and training, rather than variations in gravitational pull, which affects all swimmers equally.

3. **Answer:** B

Options: A) Niobium is a compound.; B) Niobium is an element.; C) Niobium is a metal.; D) Niobium is a mixture.

Note: The chemical symbol Nb indicates that niobium is a distinct chemical element recognized in the periodic table.

4. **Answer:** A

Options: A) once; B) twice; C) 24 times; D) 365 times

Note: The Earth completes one full rotation on its axis approximately every 24 hours, which defines the length of a day.

5. **Answer:** B

Options: A) kidney; B) skin; C) liver; D) heart

Note: The skin of a frog plays a crucial role in respiration by allowing gas exchange directly with the environment, similar to how lungs in birds facilitate oxygen intake and carbon dioxide expulsion.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Rubber is a poor conductor of heat, which helps prevent the handle of a frying pan from becoming hot and allows for safer handling while cooking.

7. **Answer:** True

Options: A) True; B) False

Note: Paramecia are unicellular organisms that can undergo conjugation, a process

where two paramecia temporarily join and exchange genetic material, thereby increasing genetic diversity.

8. **Answer:** True

Options: A) True; B) False

Note: Both burning coal and exploding fireworks involve chemical reactions that release energy in the form of heat and light, demonstrating the principle of energy transformation during combustion and rapid oxidation processes.

9. **Answer:** True

Options: A) True; B) False

Note: Organizing plant growth data by date, plant height, and soil type allows for clear comparisons and analysis of how different conditions affect plant growth over time.

10. **Answer:** False

Options: A) True; B) False

Note: Rain cannot occur at temperatures below freezing because liquid water droplets would freeze into ice or snow before reaching the ground.

Long Form Response

11. **Answer:** The label of an object can change while its velocity remains constant due to variations in perspective or reference points. For instance, a car traveling at a steady speed of 60 miles per hour can be labeled as "moving" when viewed from the roadside but may be labeled as "stationary" from inside another vehicle traveling alongside it at the same speed. This scenario illustrates how motion is relative and depends on the observer's frame of reference. Understanding this concept is important in science because it helps clarify how speed and motion are perceived differently, emphasizing that labels can be influenced by context rather than changes in actual velocity.

Note: correct because it demonstrates that the perception of motion is relative and can change based on the observer's frame of reference, illustrating how an object's label can differ while its velocity remains constant.

12. **Answer:** To calculate the momentum of an 8.0 kg bowling ball moving at 2.0 m/s, we use the formula for momentum, which is momentum (p) = mass (m) $\times$ velocity (v). Substituting the values, we have $p = 8.0 \text{ kg} \times 2.0 \text{ m/s}$, which equals $16.0 \text{ kg} \cdot \text{m/s}$. Momentum is influenced by both the mass of the object and its velocity; greater mass or higher speed results in more momentum. In real-world applications, understanding momentum is crucial for safety in sports, vehicle collisions, and even in designing safer environments, as it helps predict how objects will behave when they collide or move.

Note: correct because it accurately applies the momentum formula ($p = m \times v$) to calculate the bowling ball's momentum and discusses how both mass and velocity influence momentum, emphasizing its relevance in real-world contexts like sports and safety.

End of Answer Key